

n/1+20/1

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Economics (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is demand? What are the factors affecting demand of a commodity? Describe the law of demand with illustration. 7
- b) You deposited Rs.20,000 in your bank account at 31 December, 2006, if the bank pays 7% simple interest, how much will you accumulate in your account at 31 December 2016? What will be this amount if the bank pays compound interest? 8

OR

A truck whose price is Rs.18,600, is being paid for in 36 uniform monthly installments, including interest at 10%. After making 13 payments, the owner decides to pay off the remaining balance of the purchase price in one lump sum. How much is the lump sum?

2. a) Describe briefly the various elements and types of cost. Explain marginal cost, opportunity cost and sunk cost with suitable examples. 7
- b) Perform sensitivity analysis of the following project over a range of $\pm 20\%$ in (a) initial investment (b) salvage value (c) useful life. 8

Initial investment= 10,000

Net annual revenue= 3,000

Salvage value= 1,500

Useful life= 5 years

MARR=12%

Draw also the sensitivity graph

OR

If cost of 20-watt CFL bulb is Rs. 200 whereas cost of 100-watt filament bulb is Rs. 25, but these bulbs have equal lighting power. Which bulb do you prefer in your house? Why? Electricity cost is Rs. 15 per unit (Kw-hour).

3. a) A company purchases a piece of construction equipment for rental purposes. The expected income is Rs.3,100 annually for its useful life of 15 years. Expenses are estimated to be Rs.355 annually. If the purchase price is Rs.25,000 and there is salvage value of Rs.1500, what is the IRR? Is this equipment feasible? Also draw U/B diagram when MARR is 8%. 8
- b) A Machine costs Rs,1200000 now and its useful life is 5 years. Its salvage value is expected to be Rs.500000. Calculate depreciation in each year and corresponding book value using sinking fund method and double declining balance method. 7
4. a) A project has the following cash flow. Assume MARR = 18%. Determine the External Rate of Return (ERR) assuming reinvestment rate = 14% and calculate the simple payback period and discounted payback period. 8

EOY	NCF
0	-180,000
1	+40,000
2	+55,000
3	+65,000
4	+85,000
5	-30,000
6	+50,000

- b) Using co-terminated assumption recommend the best project taking study period as 5 years. 7

Project	A	B
Initial Investment (Rs)	3,50,000	5,00,000
Annual Revenues (Rs)	1,30,000	1,75,000
Annual cost (Rs)	15,000	25,000
Salvage value (Rs)	35,000	50,000
Useful life	5 years	8 years
MARR	10%	

5. a) Find benefit cost ratio by both methods using PW and AW method where initial investment Rs.90,000; annual income Rs.50,000; annual cost Rs.2,000; salvage value Rs.20,000 : MARR = 10% and N = 10 year. 8
- b) What do you mean by replacement analysis? Determine the choice between defender & challenger with the following information from 7

annual equivalent cost (AEC) approach when useful life is 5 years and MARR is 12%.

Item	Defender	Challenger
Initial Investment	25,00,000	35,00,000
Annual cost	10,00,000	7,50,000
Salvage value	5,00,000	12,00,000

6. a) What do you understand by capital budgeting? Explain the different methods of financing an engineering project with its advantages and disadvantages. 7
- b) What is inflation? Write down the effects of inflation in different sectors. 8
7. Write short notes on: (Any two) 2×5
 - a) Scenario Analysis
 - b) Independent, Dependent and mutually exclusive projects
 - c) Nominal Interest rate and effective interest rate